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Fine-Tuning Impairs the Balancedness of Foundation Models in Long-tailed Personalized Federated Learning

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Abstract

*Personalized federated learning (PFL) with foundation models has emerged as a promising paradigm enabling clients to adapt to heterogeneous data distributions. However, real-world scenarios often face the co-occurrence of non-IID data and long-tailed class distributions, presenting unique challenges that remain underexplored in PFL. In this paper, we investigate this long-tailed personalized federated learning and observe that current methods suffer from two limitations: (i) Fine-tuning degrades performance below zero-shot baselines due to the erosion of inherent class balance in foundation models; (ii) Conventional personalization techniques further transfer this bias to local models through parameter or feature-level fusion. To address these challenges, we propose **Fed**erated Learning via Gradient **Pu**rification and **Re**sidual **L**earning (FedPuReL), which preserves balanced knowledge in the global model while enabling unbiased personalization. Specifically, we purify local gradients using zero-shot predictions to maintain a class-balanced global model, and model personalization as residual corrections atop the frozen global model. Extensive experiments demonstrate that FedPuReL consistently outperforms state-of-the-art methods, achieving superior performance on both global and personalized models across diverse long-tailed scenarios. The code is available at* https://github.com/shihaohou/FedPuReL.

1. Introduction

Personalized federated learning (PFL) has emerged as a promising solution to address data heterogeneity in collaborative learning while preserving privacy. Recently, with the rise of pre-trained foundation models such as CLIP [35], parameter-efficient fine-tuning (PEFT) has made it viable to adapt these powerful models in federated settings, which updates a limited number of parameters while keeping the remaining ones frozen [3, 24, 48]. Recent work [8, 13, 22, 33] has demonstrated the effectiveness of this paradigm through personalized parameter mechanisms that construct separate global and local trainable parameters, leveraging aggregated global knowledge as a bridge to enhance client-specific adaptation.

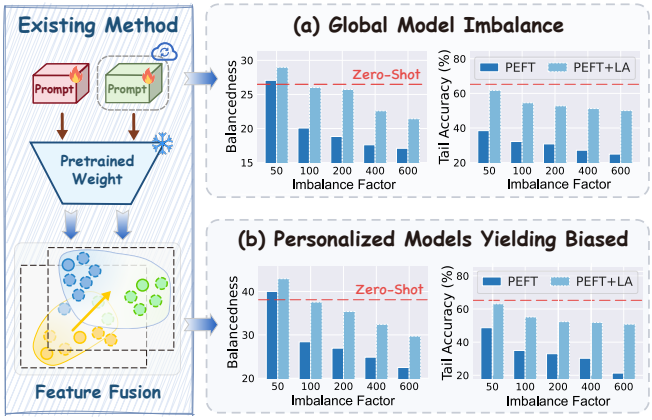


Figure 1. **Imbalanced Global Model Compromises Personalization.** We use the *Balancedness metric* [19] to measure performance distribution across classes (detailed in Sec. 3.2). Under the Fed-LT scenario, we observe that **(a)** aggregating global parameters damages the foundation model's balanced knowledge, degrading below zero-shot baselines, while **(b)** feature fusion transfers this imbalance to personalized models, yielding similarly biased predictions. Even Logit Adjustment (LA) [32], which leverages class priors, fails to restore zero-shot balancedness.

However, existing PEFT-based PFL methods have primarily focused on statistical heterogeneity under the assumption of balanced data distributions. In reality, federated learning scenarios often encounter non-IID data and long-tailed class distributions, termed federated long-tailed learning (Fed-LT) [17, 23, 46], where scarce tail-class sam-

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ples are dispersed across heterogeneous clients. This co-occurrence causes aggregated global models to become heavily biased toward head classes, hindering effective learning of tail-class representations [16, 39]. While current PEFT-based PFL methods [13, 22] demonstrate effectiveness in handling heterogeneity, they fail to preserve the balanced knowledge inherent in pre-trained foundation models when facing such long-tailed distributions.

Conversely, existing Fed-LT methods [9, 16, 36, 40] primarily focus on learning a single balanced global model through techniques such as loss re-balancing or knowledge distillation, but overlook client-specific variations. Notably, local data distributions can diverge significantly from the global long-tailed pattern, with clients potentially having different tail classes or varying degrees of imbalance.

Motivated by these limitations, our work investigates how to better leverage foundation models for long-tailed personalized federated learning. Through empirical analysis, as illustrated in Fig. 1, we identify two critical challenges that recent approaches fail to address. For the global model, naive fine-tuning significantly degrades its balancedness, leaving it *even less balanced than the original zero-shot model* in severely long-tailed scenarios. This raises our first question: **I)** ***Balancedness: How can we learn more balanced global models in Fed-LT by leveraging rather than losing the original balanced knowledge of foundation models?*** For personalized models, existing methods achieve personalization through parameter or feature-level fusion, which inevitably propagates the imbalance of the global to personalized models, yielding similarly biased predictions. Thus, this motivates our second question: **II)** ***Personalization: How can we enable effective client personalization without inheriting global bias?***

To address these questions, we develop two key insights. For I), we leverage the balanced characteristics of zero-shot CLIP predictions to constrain local updates, regularizing them toward the pre-trained model's equilibrium. This enables clients to adapt to downstream tasks without introducing head-class bias. In response to II), we reconceptualize personalization as residual learning in the output space, where client-specific adaptations become additive logit corrections applied to a frozen global model. This decouples personalization from global parameters, allowing local residual branches to capture client-specific deviations while preserving the global model's balanced representations. Building on these insights, we propose **Fed**erated Learning via Gradient **Pu**rification and **Re**sidual **L**earning (FedPuReL), which preserves balanced knowledge in the global model while enabling unbiased personalization. Our principal contributions are summarized as follows:

❶ We reveal that fine-tuning in Fed-LT inevitably erodes their inherent balanced knowledge, leading to significant bias and diminished tail-class performance.

❷ We propose FedPuReL, a balance-preserving personalization framework that constrains local gradients via zero-shot guidance to maintain a class-balanced global model while enabling residual learning for unbiased client-specific adaptations.

❸ We demonstrate superior and consistent performance improvements over strong baselines across multiple datasets with varying imbalance factors, surpassing even long-tailed strategies that leverage class priors.

2. Related Work

2.1. Federated Long-Tailed Learning

Federated long-tailed learning addresses class imbalance in FL environments, where the global long-tailed distribution intensifies statistical heterogeneity and biases global models toward head classes [15, 25, 30, 38, 51]. Recent studies have mainly pursued the following directions: (1) Loss-based rebalancing methods work by modifying loss functions to down-weight frequent classes [9, 36, 47]; (2) Classifier decoupling approaches separate representation learning from classifier training to achieve balanced optimization on the server [39]; (3) Gradient-based balancing techniques employ self-adjusting mechanisms or client grouping strategies to achieve class-wise equilibrium through accumulated gradient analysis [45, 50]. However, these methods focus solely on learning a single balanced global model, overlooking the diverse needs of individual clients. In practice, local data distributions can diverge significantly from the global long-tailed pattern. Clients may have different tail classes or varying imbalance degrees, necessitating personalized adaptations beyond the global model.

2.2. Personalized Federated Learning

Personalized federated learning extends traditional FL by developing client-specific models that adapt to local data distributions while benefiting from collaborative global knowledge [1, 18, 43]. This approach effectively addresses data heterogeneity through various strategies: regularization-based methods balance local adaptation with global consistency via proximal terms [26, 42], while model decomposition approaches separate shared representations from personalized components [2, 11, 37]. However, these methods implicitly assume balanced datasets and fail to address the compounded challenges when both non-IID and class imbalance coexist. While recent work like FedLoGe [46] attempts to integrate personalization with long-tailed learning through neural collapse, existing approaches have yet to leverage the balanced knowledge inherent in foundation models to tackle this dual challenge.

2.3. Federated Learning with Foundation models

Leveraging foundation models in federated learning has emerged as a promising direction to enhance performance

and efficiency by fine-tuning large pre-trained models rather than training from scratch. This paradigm shift addresses communication bottlenecks through parameter-efficient fine-tuning (PEFT) techniques that transmit and update only small parameter subsets. Prompt-based methods represent a major thread in this area [14], enabling clients to learn lightweight prompt parameters with subsequent work incorporating personalization strategies [13, 33] and other techniques [8, 22, 29] to handle client heterogeneity. Adapter and low-rank adaptation strategies offer complementary approaches through fine-tuning lightweight adapter modules or selectively aggregating low-rank matrices [12, 28]. Despite these advances in addressing data heterogeneity and parameter efficiency, existing foundation-model-based FL methods do not explicitly tackle class imbalance arising from long-tailed federated distributions. This gap motivates our work to integrate the representational strengths of pre-trained models with targeted mechanisms for long-tail balancing and personalization.

3. Methodology

3.1. Preliminaries

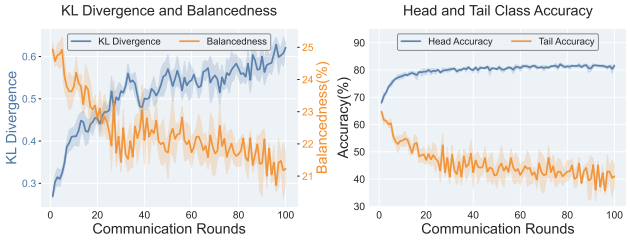
Federated Learning with Long-Tailed Data. We consider a FL framework with K clients coordinated by a central server. Each client $k \in \{1, 2, \ldots, K\}$ holds a local dataset $\mathcal{D}_k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_k}$. At each communication round t, a subset $\mathcal{S}_t \subseteq \{1, 2, \ldots, K\}$ of clients is selected to participate in training. For non-IID distributions, heterogeneity among clients is modeled using a Dirichlet distribution, where a smaller α_{dir} indicates higher client heterogeneity, demonstrating greater divergence between local and global distributions. For long-tailed distributions, the global dataset $\mathcal{D} = \bigcup_{k=1}^{K} \mathcal{D}_k$ follows a class-imbalanced distribution with class frequencies sorted as $n_1 \geq n_2 \geq \ldots \geq n_C$, where $n_c = \sum_{k=1}^{K} n_c^k$ denotes the total number of samples in class c. The degree of imbalance is defined as $\text{IF} = n_1/n_C$.

CLIP and Parameter-Efficient Fine-Tuning. We build upon CLIP [35], which performs zero-shot classification by computing cosine similarity between image and text embeddings of class prompts. To adapt CLIP to downstream tasks while preserving its zero-shot generalization, we employ PEFT methods that introduce learnable parameters $\boldsymbol{\phi}$ while freezing the pre-trained parameters $\boldsymbol{W}$.

3.2. A Closer Look at the Zero-Shot Model

To investigate the role of zero-shot foundation models in federated long-tailed learning, we conduct a preliminary analysis examining how fine-tuning affects the model's predictions and class balance. We employ two key metrics to characterize this evolution.

Temperature-Aligned KL Divergence. We introduce temperature-aligned KL divergence to quantify the structural distributional shift between fine-tuned and zero-shot predictions while neutralizing differences in prediction confidence. Standard KL divergence captures all distributional deviations without distinguishing their nature; for instance, fine-tuning on balanced data also shifts predictions, but this shift reflects improved confidence rather than bias. Our temperature-aligned metric normalizes logit scales to isolate structural bias across classes.



(a) (b)

Figure 2. **Balance degradation during federated fine-tuning.** **(a)** TKL divergence and balancedness exhibit strong negative correlation, indicating that diverging from zero-shot predictions degrades model balance. **(b)** Head-class accuracy increases while tail-class accuracy degrades, revealing amplified head-class bias.

For a sample $\mathbf{x}$ with fine-tuned logits $\mathbf{l} \in \mathbb{R}^C$ and zero-shot logits $\mathbf{z} \in \mathbb{R}^C$, we define the temperature-softmax as $\sigma_\tau(\mathbf{a}) = \text{softmax}(\mathbf{a}/\tau)$ and the entropy function as $H(\mathbf{p}) = -\sum_{c=1}^{C} p_c \log p_c$. We align both distributions to a common target entropy $H^\star = \frac{1}{2}\left[H(\sigma_1(\mathbf{l})) + H(\sigma_1(\mathbf{z}))\right]$ by computing the aligned temperatures:

$$\tau_a = H_a^{-1}(H^\star), \quad \text{where} \quad H_a(\tau) = H(\sigma_\tau(\mathbf{a})). \tag{1}$$

The temperature-aligned KL divergence is then defined as:

$$\begin{aligned} \text{TKL}(\mathbf{x}) &= D_{\text{KL}}\left(\sigma_{\tau_f}(\mathbf{f}) \,\middle\|\, \sigma_{\tau_z}(\mathbf{z})\right) \\ &= \sum_{c=1}^{C} \sigma_{\tau_f}(\mathbf{f})_c \log \frac{\sigma_{\tau_f}(\mathbf{l})_c}{\sigma_{\tau_z}(\mathbf{z})_c}. \end{aligned} \tag{2}$$

This metric neutralizes global confidence differences, preserving focus on class bias.

Balancedness Metric. We adopt a balancedness metric, introduced in [19], aiming to measure how balanced the model is by computing performance equity across classes:

$$\beta(V) = \frac{1}{C^2} \sum_{i,j} \exp\left(-\frac{|a_i - a_j|^2}{\sigma}\right), \tag{3}$$

where a_i denotes the accuracy of class i and σ is a fixed scaling parameter. This metric reaches its maximum value of 1 when all classes achieve equal accuracy, directly quantifying cross-class performance equity.

Using these two metrics, we analyze the fine-tuning dynamics of CLIP on federated long-tailed scenarios. Specifically, we track both TKL divergence and balancedness

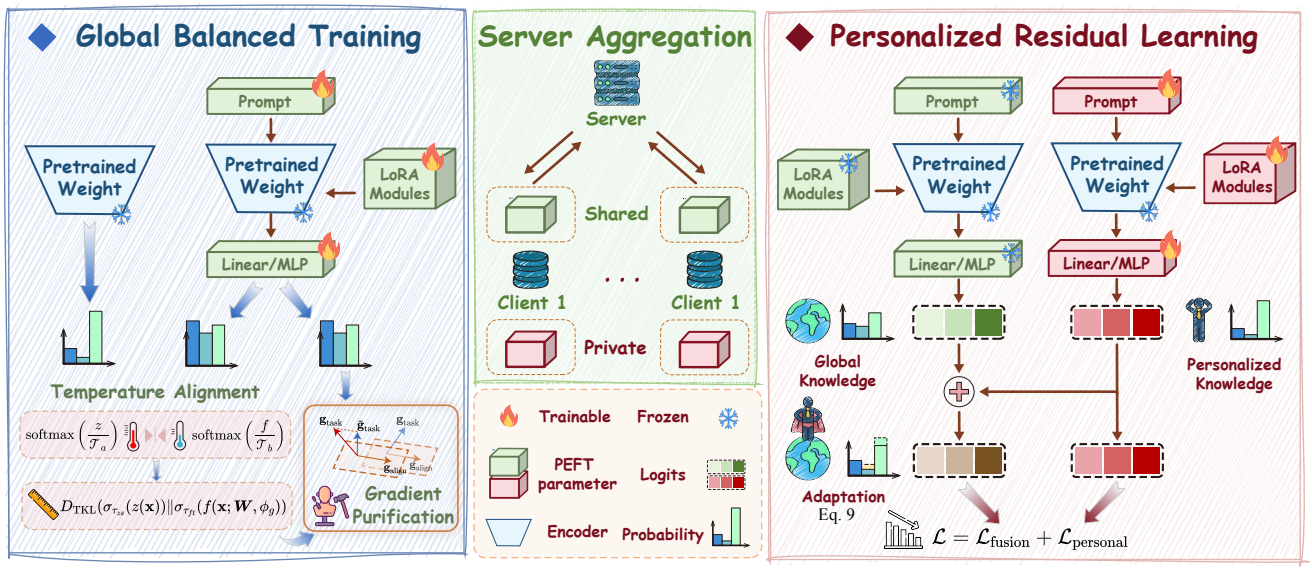


Figure 3. **Overview of FedPuReL framework.** (Left) *Global Balanced Training:* Clients train shared PEFT parameters (e.g., LoRA modules) using zero-shot guided gradient alignment. Temperature-aligned predictions enable gradient purification via projection. (Center) *Server Aggregation:* Shared parameters are aggregated; personalized parameters remain private. (Right) *Personalized Residual Learning:* Personalization stage learns client-specific residuals over the frozen global parameters through dual-branch additive fusion.

$\beta(V)$ across training rounds to examine how fine-tuning affects the model's predictions and class balance. Our analysis reveals two critical patterns:

Observation 1: Negative correlation between divergence and balance. As shown in Fig. 2(a), TKL divergence and balancedness $\beta(V)$ exhibit a strong negative correlation across training rounds. As fine-tuning progresses, predictions increasingly deviate from the zero-shot model, while class balance concurrently deteriorates. This reveals that adapting to long-tailed downstream tasks gradually erodes the balanced knowledge inherent in the pre-trained model.

Observation 2: Amplified head-class bias. Fig. 2(b) shows that fine-tuning disproportionately benefits head classes while degrading tail-class performance, significantly widening the performance gap. This asymmetric adaptation actively amplifies distributional bias, directly contributing to the declining balancedness.

These observations reveal that fine-tuning foundation models on long-tailed data creates a fundamental trade-off between task adaptation and preserving the balanced knowledge inherent in zero-shot predictions. Pre-trained CLIP models possess robust cross-class equilibrium from large-scale diverse training, but this equilibrium deteriorates as they adapt to imbalanced downstream distributions.

3.3. Proposed Method

We propose FedPuReL to learn PEFT parameters that adapt foundation models to downstream tasks while preserving their inherent balanced representations, as depicted in Fig. 3. We employ gradient purification to maintain global balance during collaborative training, and residual learning to achieve unbiased client-specific adaptation. This ensures task adaptation without amplifying class imbalances.

3.3.1. Balancedness: Gradient Purification

As demonstrated in Sec. 3.2, naive fine-tuning inevitably shifts the model away from balanced zero-shot representations, causing performance degradation on tail classes. To address this, we constrain local gradient updates to directions that do not deviate from the zero-shot model, thereby preserving its balanced knowledge while adapting to downstream tasks. Specifically, we encourage the fine-tuned model to remain aligned with the balanced distribution of the zero-shot model through TKL divergence:

$$\mathcal{L}_{\text{align}} = D_{\text{TKL}}\left(\sigma_{\tau_{zs}}(\mathbf{z}(\mathbf{x})) \,\middle\|\, \sigma_{\tau_{ft}}(\mathbf{f}(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_g))\right), \quad (4)$$

where $\mathbf{z}$ and $\mathbf{f}$ denote the zero-shot and fine-tuned logits, respectively, and τ_{zs}, τ_{ft} are temperature parameters controlling the smoothness of the softmax distributions. The gradient $\mathbf{g}_{\text{align}} = \nabla_{\boldsymbol{\phi}_g}\mathcal{L}_{\text{align}}$ points toward preserving the balanced knowledge encoded in the zero-shot model.

To preserve this balanced knowledge, we project the gradient onto the subspace orthogonal to $\mathbf{g}_{\text{align}}$:

$$\tilde{\mathbf{g}}_{\text{task}} = \begin{cases} \mathbf{g}_{\text{task}}, & \text{if } \langle \mathbf{g}_{\text{task}}, \mathbf{g}_{\text{align}} \rangle \geq 0 \\ \mathbf{g}_{\text{task}} - \dfrac{\langle \mathbf{g}_{\text{task}}, \mathbf{g}_{\text{align}} \rangle}{\|\mathbf{g}_{\text{align}}\|^2} \mathbf{g}_{\text{align}}, & \text{otherwise.} \end{cases} \quad (5)$$

where $\mathbf{g}_{\text{task}} = \nabla_{\boldsymbol{\phi}_g} \text{CE}(y, \sigma(\mathbf{f}(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_g)))$. When the task gradient conflicts with $\mathbf{g}_{\text{align}}$, naively applying it would push

the model away from the pre-trained anchor. This projection removes any component of $\mathbf{g}_{\text{task}}$ that opposes $\mathbf{g}_{\text{align}}$, ensuring that every update preserves balanced knowledge while adapting to downstream tasks.

After local training with the purified gradient, clients upload parameters to the server for aggregation:

$$\boldsymbol{\phi}_g^{(t+1)} \leftarrow \sum_{k \in \mathcal{S}_t} \frac{n_k}{\sum_{j \in \mathcal{S}_t} n_j} \boldsymbol{\phi}_g^{k,(t)}, \tag{6}$$

where $\boldsymbol{\phi}_g^{k,(t)}$ is obtained through local updates using $\tilde{\mathbf{g}}_{\text{task}}$. The resulting global model $(\boldsymbol{W}, \boldsymbol{\phi}_g^{(t+1)})$ achieves balanced, task-adapted representations across clients, providing a stable foundation for client-specific personalization.

3.3.2. Personalization: Learning Clients Residuals

To achieve personalization, recent methods fuse global and local information at the feature or parameter level to construct personalized models [22, 26, 33]. However, such fusion strategies inevitably transfer class imbalances to local models, which limits their personalization effectiveness. Instead, we model personalization as learning corrective residuals that build upon the balanced global foundation. Specifically, these residuals capture client-specific patterns as additive corrections atop the global model's balanced, task-adapted knowledge. We realize this through a dual-branch architecture with two parallel pathways:

Global Branch. This branch produces baseline predictions:

$$\mathbf{l}_{\text{G}}(\mathbf{x}) = f(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_g), \tag{7}$$

where $f(\cdot)$ denotes the forward pass through the global model. Critically, $\boldsymbol{\phi}_g$ remains frozen throughout the personalization stage, serving as an immutable anchor that preserves the balanced, task-adapted knowledge.

Personalized Branch. In parallel, each client k learns client-specific parameters $\boldsymbol{\phi}_k$ that capture local patterns:

$$\mathbf{l}_{\text{P}}^k(\mathbf{x}) = f(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_k). \tag{8}$$

The personalized parameters $\boldsymbol{\psi}_k$ can be flexibly instantiated as an independent classification head, additional adapter layers, learnable prompts, or LoRA modules.

Additive Fusion. The final prediction combines both branches additively:

$$\mathbf{l}_{\text{final}}^k(\mathbf{x}) = \underbrace{f(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_g)}_{\text{Global Part}} + \underbrace{f(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_k)}_{\text{Personal Part}}. \tag{9}$$

The personalized model inherently stands on the shoulders of giants, benefiting from the global model's balanced knowledge while adapting to local specifics. Even under local overfitting or shifts, the global branch offers a reliable anchor, averting severe degradation.

During local training, we optimize a composite loss:

$$\mathcal{L}_{\text{fusion}}^k = \text{CE}\big(y,\, \sigma(\mathbf{f}_{\text{G}}(\mathbf{x}) + \mathbf{f}_{\text{P}}^k(\mathbf{x}))\big), \tag{10}$$

$$\mathcal{L}_{\text{personal}}^k = \text{CE}\big(y,\, \sigma(\mathbf{f}_{\text{P}}^k(\mathbf{x}))\big), \tag{11}$$

$$\mathcal{L}_{\text{total}}^k = (1-\lambda)\,\mathcal{L}_{\text{fusion}}^k + \lambda\,\mathcal{L}_{\text{personal}}^k. \tag{12}$$

where λ controls the trade-off. The fusion loss $\mathcal{L}_{\text{fusion}}^k$ maintains stability by optimizing the combined prediction, while the personalization loss $\mathcal{L}_{\text{personal}}^k$ drives parameter specialization to capture local patterns. Gradients flow solely through $\boldsymbol{\phi}_k$, safeguarding global balance.

During inference, global model only uses the global branch $p(y|\mathbf{x}) = \sigma(f(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_g))$. For personalized inference, we fuse both branches: $p_k(y|\mathbf{x}) = \sigma(f(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_g) + f(\mathbf{x}; \boldsymbol{W}, \boldsymbol{\phi}_k))$. This fusion combines the robust global model with client-specific corrections, achieving superior accuracy on local distributions while maintaining the stability inherited from balanced global training.

4. Experiments

4.1. Experimental Setup

Datasets. We conduct experiments on Food101-LT [4], DTD-LT [7], FGVC-Aricraft-LT [31], Stanford Dogs-LT [20], OxfordPets-LT [34], and CIFAR-100-LT [21], ImageNet-LT [10], Places-LT [52] datasets. For the first six datasets, we follow [6] to create long-tailed versions using exponential decay with imbalance factor. For ImageNet-LT and Places-LT, we use the standard version from [27].

Evaluation Protocol. To evaluate global model (GM) performance, we calculate test accuracy using a globally balanced dataset and report accuracy across many (>100 samples), medium (20-100 samples), and few (<100 samples) classes. For personalized model (PM) evaluation, we employ local test accuracy, where each local test set is sampled from the global test set to maintain an identical distribution to its corresponding local training set. The PM accuracy is computed as the arithmetic mean of local test accuracies across all clients. Detailed categorization criteria for these class groups are provided in the Appendix.

Baselines. We compare FedPuReL against three representative PEFT categories: ❶ *Prompt-based* , ❷ *LoRA-based*, and ❸ *Adapter-based*, including: PromptFL [14], PromptFolio [33], FedOTP [22], FedPGP [8], CLIPLoRA [49], FedSA-LoRA [12] and FedClip [28].

To demonstrate FedPuReL's advantages over methods that explicitly address class imbalance, we additionally include Fed-GraB [45], which uses gradient norm-based reweighting to mitigate head-class bias, combined with strong baselines (PromptFolio+Fed-GraB, CLIPLoRA+Fed-GraB, FedClip+Fed-GraB).

Implementation Details. For a fair comparison, we follow the experimental settings used in prior work [14, 22, 28,

Table 1. **Comparison with state-of-the-art methods on ImageNet-LT and Places-LT** for global models (GM) and personalized models (PM) across all classes and shot categories (Many, Med., Few). Best in **bold**, second-best underlined.

Method	ImageNet-LT								Places-LT							
	GM				PM				GM				PM			
	All	Many	Med.	Few	All	Many	Med.	Few	All	Many	Med.	Few	All	Many	Med.	Few
Zero-shot	67.05	67.88	66.49	66.65	66.68	67.71	66.07	65.97	35.55	33.39	35.21	40.31	35.33	33.59	35.21	40.63
Prompt-based																
PromptFL [14]	67.76	76.20	63.71	58.36	67.51	75.48	63.69	58.82	38.90	51.07	32.86	29.64	36.49	48.00	31.37	27.42
PromptFolio [33]	69.64	**80.26**	66.37	47.02	67.65	77.23	64.89	45.99	40.55	52.57	36.99	31.58	37.95	47.73	34.52	28.32
FedOTP [22]	57.03	69.57	59.58	43.74	50.49	66.32	42.60	34.14	36.07	48.18	32.23	22.66	32.56	43.28	29.26	21.11
FedPGP [8]	67.05	76.99	62.99	53.60	66.44	76.59	62.32	52.97	38.92	50.82	32.89	30.27	37.36	48.21	32.04	28.90
PromptFolio+Fed-Grab [45]	69.53	79.90	66.16	52.83	68.14	**77.32**	65.09	52.11	41.99	52.82	38.43	30.82	39.14	47.82	**36.04**	30.76
FedPuReL (ours)	**72.96**	77.32	**68.41**	**66.70**	**70.12**	76.34	**67.68**	**66.62**	**43.88**	**55.44**	**40.94**	**39.06**	**39.68**	**49.85**	35.55	**32.72**
	↑ 3.32	↓ 2.94	↑ 2.04	↑ 8.34	↑ 1.98	↓ 0.98	↑ 2.59	↑ 7.80	↑ 1.89	↑ 2.62	↑ 2.51	↑ 7.48	↑ 0.54	↑ 1.64	↓ 0.49	↑ 1.96
LoRA-based																
CLIPLoRA [49]	70.18	**80.91**	65.84	52.90	66.07	**77.82**	61.77	48.86	41.61	54.51	36.49	29.79	37.21	49.10	33.14	25.30
FedSA-LoRA [12]	69.44	80.07	66.84	58.83	61.73	71.74	57.52	48.46	40.90	53.85	35.81	28.81	37.46	49.33	33.27	25.89
CLIPLoRA+Fed-Grab [45]	70.24	79.43	66.75	57.09	68.94	77.14	65.56	55.56	42.00	53.56	37.69	30.92	38.31	49.17	34.77	27.16
FedPuReL (ours)	**71.43**	78.89	**67.69**	**66.31**	**70.55**	76.09	**67.63**	**66.98**	**44.41**	**60.72**	**43.08**	**35.79**	**41.13**	**52.58**	**35.42**	**32.83**
	↑ 1.19	↓ 2.02	↑ 0.85	↑ 7.48	↑ 4.48	↓ 1.73	↑ 2.07	↑ 11.42	↑ 2.41	↑ 6.21	↑ 5.39	↑ 4.87	↑ 2.82	↑ 3.25	↑ 0.65	↑ 5.67
Adapter-based																
FedClip [28]	68.97	74.59	65.67	59.47	68.63	**75.19**	65.43	59.02	39.28	47.40	34.75	34.66	38.59	**47.13**	33.78	33.74
FedClip+Fed-Grab [45]	69.25	**75.68**	66.23	61.85	67.50	72.33	65.07	62.81	39.43	46.52	35.34	35.68	38.95	45.95	34.81	35.56
FedPuReL (ours)	**70.51**	73.52	**69.79**	**67.49**	**70.19**	74.24	**69.27**	**67.52**	**41.99**	**51.51**	**39.00**	**38.89**	**40.12**	45.48	**37.06**	**36.39**
	↑ 1.26	↓ 2.16	↑ 3.56	↑ 5.64	↑ 1.56	↓ 0.95	↑ 3.84	↑ 4.71	↑ 2.56	↑ 4.11	↑ 3.66	↑ 3.21	↑ 1.17	↓ 1.65	↑ 2.25	↑ 0.83

33]. We employ CLIP with ViT-B/16 as the backbone. For prompt-based methods, the prompt length is set to 4, while for LoRA-based methods, the LoRA rank is set to 8. In our federated learning setup, we deploy 20 clients with non-IID data distribution controlled by Dirichlet parameter $\alpha = 1$ and IF $= 100$ by default. Training is conducted over 100 communication rounds using SGD optimizer, with 40% of clients randomly selected to participate in each round.

4.2. Comparison with State-of-the-Art

Performance on Large-Scale Benchmarks. Table 1 compares FedPuReL on ImageNet-LT and Places-LT. FedPuReL consistently outperforms all baselines in both GM and PM settings, particularly on few classes, even exceeding Fed-GraB-enhanced variants. This performance pattern aligns with our design principle: gradient purification maintains zero-shot balance by preventing the global model from overfitting to head classes, instead reallocating optimization capacity toward under-represented categories. As evidenced by Fig. 4 and the analysis in Sec. 4.5, this mechanism yields substantially improved tail performance and higher balancedness compared to baseline. Table 3 demonstrates robustness across five additional datasets spanning fine-grained recognition, textures, and scenes.

Performance Under Varying Imbalance Factors. Table 2 evaluates FedPuReL on CIFAR-100-LT across varying imbalance factors. FedPuReL maintains superior performance as imbalance intensifies, with advantages growing proportionally, demonstrating effectiveness in severe long-tail scenarios. Fed-GraB's limited gains validate that relying solely on prior estimation is insufficient under severe long-tail scenarios, whereas our gradient purification naturally maintains balance by anchoring to zero-shot predictions.

Table 2. **Comparison with state-of-the-art methods** on CIFAR-100-LT under varying imbalance factors for global models (GM) and personalized models (PM).

CIFAR-100-LT Method	IF=500		IF=200		IF=100		IF=50	
	GM	PM	GM	PM	GM	PM	GM	PM
Zero-shot	64.82	64.79	64.82	64.79	64.82	64.79	64.82	64.79
Prompt-based								
PromptFL [14]	56.12	67.50	61.21	71.90	61.98	70.52	64.50	72.51
PromptFolio [33]	57.36	63.80	60.58	69.74	64.10	70.54	67.15	72.69
FedOTP [22]	52.59	58.43	51.33	65.09	58.02	65.23	61.30	68.64
FedPGP [8]	55.88	62.69	59.48	67.47	62.96	67.83	65.55	71.23
PromptFolio+Fed-GraB [45]	58.98	65.28	62.18	71.00	65.14	71.79	68.49	73.60
FedPuReL (ours)	**67.20**	**72.70**	**68.01**	**74.67**	**69.77**	**73.37**	**70.40**	**74.68**
	↑ 9.84	↑ 5.20	↑ 5.83	↑ 2.77	↑ 4.63	↑ 1.58	↑ 1.91	↑ 1.08
LoRA-based								
CLIPLoRA [49]	69.37	73.68	71.71	76.63	75.02	77.80	76.85	79.52
FedSA-LoRA [12]	66.51	74.26	71.66	76.50	74.48	81.68	72.86	78.61
CLIPLoRA+Fed-GraB [45]	70.32	74.17	72.58	77.43	76.35	77.91	77.47	80.83
FedPuReL (ours)	**72.91**	**77.73**	**74.26**	**80.04**	**77.56**	**84.62**	**78.12**	**81.85**
	↑ 2.59	↑ 3.47	↑ 1.68	↑ 2.61	↑ 1.21	↑ 2.94	↑ 0.65	↑ 1.02
Adapter-based								
FedClip [28]	61.49	65.77	62.68	68.61	64.09	66.87	65.59	68.68
FedClip+Fed-GraB [45]	62.51	67.14	63.86	69.48	65.40	67.20	66.60	**69.99**
FedPuReL (ours)	**66.52**	**68.44**	**66.55**	**69.84**	**67.54**	**68.42**	**67.13**	69.89
	↑ 4.01	↑ 1.30	↑ 2.69	↑ 0.36	↑ 2.14	↑ 1.22	↑ 0.53	↓ 0.10

4.3. Ablation Study

To thoroughly assess our approach, we conduct ablation studies investigating the efficacy of each component under long-tailed scenarios. We perform ablation experiments on CIFAR-100-LT and Places-LT datasets.

Table 3. **Comparison with state-of-the-art methods** on Food101-LT, DTD-LT, Aircraft-LT, Dogs-LT, and Pets-LT datasets.

Method	Food101-LT		DTD-LT		Aircraft-LT		Dogs-LT		Pets-LT	
	GM	PM	GM	PM	GM	PM	GM	PM	GM	PM
Zero-shot	85.39	85.53	48.41	41.46	21.54	29.38	66.43	60.19	90.41	86.03
Prompt-based										
PromptFL [14]	82.74	88.06	46.22	63.49	26.59	28.51	62.57	67.61	86.80	91.98
PromptFolio [33]	81.49	87.09	54.97	70.02	29.26	34.68	64.94	70.30	85.15	91.08
FedOTP [22]	74.61	84.52	48.70	68.63	22.50	24.55	64.76	65.17	67.46	83.73
FedPGP [8]	81.43	85.91	48.47	64.96	24.81	26.70	62.44	65.85	84.85	92.00
PromptFolio+Fed-GraB [45]	83.59	88.03	56.21	70.21	29.91	**35.12**	63.52	61.03	87.73	91.96
FedPuReL (ours)	**86.71**$_{\uparrow 3.12}$	**89.26**$_{\uparrow 1.20}$	**57.75**$_{\uparrow 1.54}$	**70.46**$_{\uparrow 0.25}$	**32.81**$_{\uparrow 2.90}$	34.15$_{\downarrow 0.97}$	**67.06**$_{\uparrow 2.12}$	**72.48**$_{\uparrow 2.18}$	**92.13**$_{\uparrow 4.40}$	**94.53**$_{\uparrow 2.57}$
LoRA-based										
CLIPLoRA [49]	82.27	87.43	51.76	66.83	30.02	33.61	67.49	71.28	88.77	93.56
FedSA-LoRA [12]	83.40	86.50	44.53	68.60	26.31	31.23	66.10	67.08	88.05	93.71
CLIPLoRA+Fed-GraB [45]	84.84	87.69	54.00	69.07	31.62	33.35	68.29	69.63	89.29	93.91
FedPuReL (ours)	**86.63**$_{\uparrow 1.78}$	**89.32**$_{\uparrow 1.63}$	**54.45**$_{\uparrow 0.45}$	**69.56**$_{\uparrow 0.49}$	**32.68**$_{\uparrow 1.06}$	**37.02**$_{\uparrow 3.41}$	**69.25**$_{\uparrow 0.96}$	**73.54**$_{\uparrow 2.26}$	**91.98**$_{\uparrow 2.69}$	**94.60**$_{\uparrow 0.69}$
Adapter-based										
FedClip [28]	83.75	85.76	46.01	50.47	22.83	22.66	63.74	61.10	89.24	90.55
FedClip+Fed-GraB [45]	84.35	86.14	44.60	49.33	22.59	22.78	63.52	61.03	90.01	91.63
FedPuReL (ours)	**85.76**$_{\uparrow 1.41}$	**87.08**$_{\uparrow 0.94}$	**48.15**$_{\uparrow 2.14}$	**51.52**$_{\uparrow 1.05}$	**24.78**$_{\uparrow 1.95}$	**24.16**$_{\uparrow 1.38}$	**66.01**$_{\uparrow 2.27}$	**62.10**$_{\uparrow 1.00}$	**90.98**$_{\uparrow 0.97}$	**92.76**$_{\uparrow 1.13}$

Table 4. **Comparison on CIFAR100 and Places-LT** under Global (G) and Personalized (P) settings. Best in **bold**, second-best underlined.

Method	G	P	CIFAR-100-LT								Places-LT							
			GM				PM				GM				PM			
			All	Many	Med.	Few	All	Many	Med.	Few	All	Many	Med.	Few	All	Many	Med.	Few
Baseline	✗	✗	61.98	80.23	61.12	42.32	70.52	82.53	62.15	48.67	38.90	51.07	32.86	29.64	36.49	48.00	31.37	27.42
+ Gradient Purification	✓	✗	69.77	78.63	67.08	64.74	72.06	81.91	63.64	57.46	43.88	55.44	40.94	39.06	38.14	48.34	35.27	32.72
v1. LS-KD [41]	✓	✗	66.43	80.77	64.29	52.58	71.56	81.59	63.32	55.26	40.42	50.58	34.34	35.49	37.47	48.22	33.95	32.89
v2. WiSE-FT [44]	✓	✗	63.64	78.86	62.03	49.42	69.94	80.92	62.48	50.11	38.95	48.67	33.72	39.14	37.04	44.81	32.52	32.90
+ Residual	✗	✓	68.43	78.80	65.82	59.58	72.25	82.28	63.30	55.89	42.27	49.84	37.25	39.64	39.34	48.75	33.88	32.35
v1. PromptFolio [33]	✗	✓	65.90	83.74	66.82	44.74	69.91	84.93	62.66	46.49	41.30	52.90	36.47	31.01	37.14	47.69	34.20	28.58
v2. FedOTP [22]	✗	✓	62.96	84.03	63.72	41.68	66.53	83.36	58.3	35.49	36.37	48.73	32.27	23.07	34.25	46.03	30.41	22.12
Ours	✓	✓	**69.77**	**78.63**	**67.08**	**64.74**	**73.37**	**82.67**	**65.69**	**58.25**	**43.88**	**55.44**	**40.94**	**39.06**	**39.68**	**49.85**	**34.55**	**35.56**

v1/v2: Contender for initialization and regularization components.

Balancing Strategies for Global Training. Starting from a baseline without balancing mechanisms, we compare our method against alternative zero-shot guidance approaches:

- *LS-KD* [41]: Apply Z-score standardization to logits before computing softmax and KL divergence.
- *WiSE-FT* [44]: Ensemble zero-shot and fine-tuned models in weight space by linearly interpolating with mixing coefficient $\alpha \in [0, 1]$ to improve OOD robustness.

As shown in Table 4, our *Gradient Purification* achieves superior performance, demonstrating that projecting task gradients orthogonal to alignment gradients effectively preserves CLIP's balanced knowledge during task adaptation.

Personalization Strategies. We evaluate our residual learning against existing personalized fusion approaches:

- *PromptFolio* [33]: Employ portfolio-based prompt selection via feature-level interpolation.
- *FedOTP* [22]: Employ unbalanced optimal transport to align local visual features with global and local prompts.

Our *Residual Learning*, even without gradient purification, surpasses both alternatives. This demonstrates that output-space residuals effectively mitigate bias inheritance.

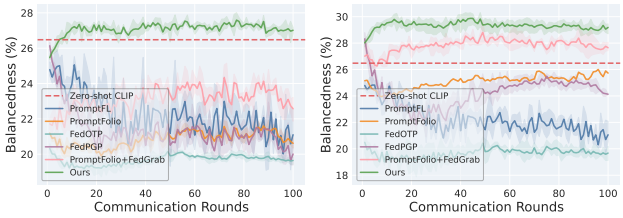
Integrating gradient purification with residual learning yields gains exceeding those of individual components, demonstrating their synergy. The combination achieves the best results across both datasets. These findings confirm our thesis that jointly addressing balancedness and personalization is essential for effective federated long-tailed learning.

4.4. More Supporting Experiments

Due to the page limitation, we report additional results and ablation studies in Appendix D, including 1) different client heterogeneities; 2) ablation hyper-parameter λ in Eq. 12; 3) analysis of gradient angles; 4) model convergence.

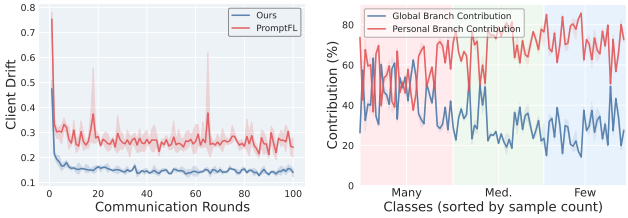
4.5. Analysis of FedPuReL

Analysis 1: FedPuReL effectively maintains class balance. To assess whether FedPuReL preserves balance, we track the balancedness metric $\beta(V)$ throughout training (Fig. 4). Global models trained with standard FL methods exhibit significant balance degradation over time. In contrast, FedPuReL maintains close to the initial zero-shot level, demonstrating effective balance preservation through



(a) Global Model (b) Personalized models

Figure 4. **Comparison of Balancedness** of SOTA prompt-based methods on global and personalized models for CIFAR-100-LT.



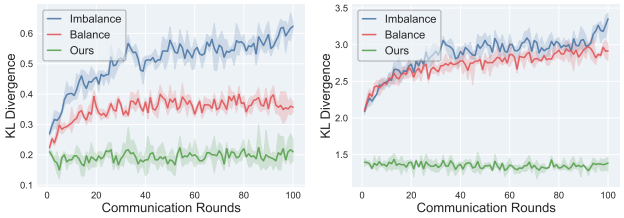
(a) Client drift evolution (b) Branch contribution

Figure 5. **Client drift and branch contribution analysis. (a)** Client drift measured as the L2 distance between local and global model. **(b)** Percentage contribution of global versus personalized branches to correct predictions, decomposed across classes sorted. Both analyses are conducted on CIFAR-100-LT.

gradient purification. For personalized models, standard approaches inherit and propagate global bias, resulting in poor balancedness. FedPuReL substantially improves balancedness, validating that residual learning enables client-specific personalization without bias propagation. The minimal gap between global and personalized balancedness in FedPuReL, demonstrates that our approach uniformly benefits all clients across diverse data distributions, rather than trading global fairness for personalization gains.

Analysis 2: TKL metric quantifies divergence from zero-shot knowledge. We employ our TKL metric, which neutralizes confidence variations to isolate true distributional changes (Fig. 6). Standard KL divergence exhibits large values even on balanced data due to confidence shifts, obscuring genuine bias. In contrast, TKL distinguishes balanced adaptation from biased divergence. Comparing results across datasets reveals that TKL divergence is higher on long-tailed data, quantifying how class imbalance systematically drives models away from CLIP's balanced knowledge. The strong negative correlation between TKL and $\beta(V)$ (Fig. 2) confirms this relationship.

Analysis 3: FedPuReL effectively reduces client drift. Intuitively, when different clients purify their gradients toward the same zero-shot anchor, this shared reference point should improve cross-client consistency. To evaluate this, we measure client drift as the L2 distance between local and global parameters: $\text{drift}_k^{(t)} = \|\theta_k^{(t)} - \theta_{\text{global}}^{(t)}\|_2$. As illustrated in Fig. 5a, FedPuReL maintains substantially lower



(a) Temperature-aligned KL (b) KL

Figure 6. **Comparison of different KL divergence metrics** across imbalanced and balanced CIFAR-100 datasets.

drift than PromptFL with minimal variance. In contrast, PromptFL shows volatile spikes, suggesting episodic severe divergence where some clients' updates conflict sharply with the global model. These results demonstrate that gradient purification maintains cross-client consistency with the global model, enabling faster convergence through improved aggregation quality.

Analysis 4: Global and personalized branches exhibit complementary contributions. Our design separates balanced global knowledge from personalized local adaptation, which should lead to different contribution patterns across class types. Specifically, head classes with abundant samples should rely more on global knowledge, while tail classes with scarce data should require stronger personalized corrections. To evaluate this, we decompose each correct prediction to identify the dominant contributor between the global and personalized branches. As illustrated in Fig. 5b, as class frequency decreases from head to tail, the personalized branch contribution increases, while the global branch contribution decreases correspondingly. Notably, even for tail classes where personalization dominates, the global branch maintains a substantial 20-25% contribution. These results confirm our design principle: the global branch provides consistent balanced knowledge across all classes, while the personalized branch dynamically adjusts its contribution to compensate for data scarcity.

5. Conclusion

In this work, we explored the long-tail problem in personalized federated learning based on PEFT fine-tuning of foundation models. We observe that conventional approaches erode the pre-trained model's inherent class balance and propagate bias to client-specific adaptations. We introduced FedPuReL, a framework that maintains global balance via zero-shot-guided gradient purification and supports unbiased personalization through residual learning in the output space. Extensive experiments on diverse datasets and imbalance factors showed that FedPuReL outperforms baselines for both global and personalized models, even without class priors. These results highlight how pre-trained knowledge can alleviate class imbalances in federated settings, fostering more equitable collaborative learning.

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